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Z → A = 2
Z → P = 3
A → P = .1
A → E = 1
P → M = .4
E → P = 2
E → W = 10
E → U = .5
U → Z = 3
U → W = 4

Start Z

Z = 0 {Z} {Z,A} {Z,A,P,M,E} Z A P M E

A = ~~∞~~ 2

P = ~~∞~~ ~~3~~ 2.1

M = ~~∞~~ 2.5

E = ~~∞~~ 3

W = ~~∞~~ 13

U = ~~∞~~ 3.5

Cont. 2.5 is Better

W ~~7.5~~

Final

	Dist	SP
Z	0	Z
A	2	Z → A
P	2.1	Z → A → P
M	2.5	Z → A → P → M
E	3	Z → A → E
U	3.5	Z → A → E → U
W	7.5	Z → A → E → U → W

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Algorithm Analysis

Prob 2: BFS Shortest Path

time: $O(n^2)$ because for every node we visit we have to scan n neighbors and can visit n nodes

space: $O(n)$ because we use a queue and two arrays so at most we are holding n elements

Prob 3

Time: $O(n^2)$ because we scan all nodes to find smallest unvisited distance and go through all nodes

space: $O(n)$ because we use 3 arrays that use n space total

Prob 4

Time: $O(H+K)$ where h is numCourses and e is numPairs

space: we store the list $O(H+K)$

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